NETWORK TRAFFIC ANALYSIS USING WIRESHARK

DNS, TCP and UDP Packet Inspection

Project Report

Tool: Wireshark 4.6.9
Operating System: Windows | Capture Interface: Ethernet

Prepared as a practical network traffic analysis project

1. Introduction

Network traffic analysis involves capturing and examining packets exchanged between devices. This project used Wireshark to inspect DNS communication and selected TCP and UDP traffic from a live Ethernet capture. The work focused on understanding packet fields, transport protocols, DNS records and response behaviour.

2. Objectives

• Capture live network traffic from the Ethernet interface.
• Identify DNS queries and their corresponding responses.
• Examine source and destination IP addresses and port numbers.
• Observe TCP connection establishment and termination flags.
• Inspect UDP header fields and identify UDP-based QUIC traffic.
• Record observations without assuming that ordinary traffic is malicious.

3. Tools and Environment

Component	Details
Operating system	Windows
Packet analysis tool	Wireshark 4.6.9
Capture interface	Ethernet
Capture type	Live packet capture
Capture file	wireshark_EthernetLYK7W3.pcapng (as shown in screenshot)

4. Methodology

The Ethernet interface was selected in Wireshark and live packets were captured while normal network activity was taking place. Display filters such as `dns`, `tcp` and `udp` were used to isolate traffic. Individual packets were selected and their protocol layers and fields were inspected.

5. DNS Traffic Analysis

5.1 DNS Query (Packet 38)

Packet 38 contained a DNS A-record query for **web.whatsapp.com**. The query originated from the local device and was sent to the local network DNS resolver. Wireshark identified the DNS message as being carried over TCP, with source port 60414 and destination port 53.

Field	Observed value
Packet number	38
Domain queried	web.whatsapp.com
Query type	A (IPv4 address)
Source IP	192.168.0.170
Destination IP	192.168.0.1
Transport	TCP
Source / destination ports	60414 / 53

5.2 DNS Response (Packet 47)

The matching response was packet 47. It returned a successful DNS response with the status **No error** and contained two answer records: a CNAME record and an A record. Wireshark linked the response to request packet 38 and reported a response time of approximately 6.2757 milliseconds.

Field	Observed value
Response packet	47
Response code	No error
Answer records	2
CNAME	mmx-ds.cdn.whatsapp.net
A record	57.144.211.32
CNAME TTL	495 seconds
A record TTL	55 seconds
Response time	6.2757 ms

5.3 TTL Interpretation

Time to Live (TTL) indicates how long a DNS record may remain cached. The observed CNAME TTL was 495 seconds and the A-record TTL was 55 seconds. Different TTL values for different DNS records are normal and are not, by themselves, evidence of a security issue.

6. TCP Traffic Analysis

TCP packets were examined using the `tcp` display filter. The capture showed a three-way handshake for a connection between the local device (192.168.0.170) and the DNS resolver (192.168.0.1), using client port 51990 and server port 53. Packets 29–31 showed SYN, SYN-ACK and ACK, respectively. FIN/ACK packets were also observed later, indicating graceful connection termination.

Packet	Observed TCP flags	Meaning
29	SYN	Client initiates connection
30	SYN, ACK	Server acknowledges and responds
31	ACK	Client acknowledges; handshake completes

7. UDP Traffic Analysis

UDP packets were inspected using the `udp` display filter. One selected packet showed communication from the local device to a remote host on destination port 443. QUIC traffic was also visible in the capture; QUIC commonly uses UDP as its transport protocol.

Field	Observed value
Packet number	8
Source IP	192.168.0.170
Destination IP	142.251.155.119
Source port	57921
Destination port	443
UDP length	37 bytes
UDP payload	29 bytes

8. Observations and Security Assessment

The capture demonstrated normal DNS resolution, TCP connection establishment and termination, and UDP-based communication associated with QUIC. The examined DNS response returned 'No error'. No malicious activity or vulnerability can be concluded from these selected packets alone. This project was a basic protocol and packet inspection exercise, not a comprehensive network security audit.

9. Recommendations

• Continue monitoring DNS requests and investigate unexpected or repeated failures.
• Review unusual outbound connections and unexpected destination ports.
• Use secure network configurations and trusted DNS resolvers.
• Preserve packet captures securely and remove unrelated personal traffic before sharing.
• Correlate suspicious packet patterns with endpoint and network logs before classifying activity as malicious.

10. Conclusion

The project provided practical experience in capturing and analysing live network traffic with Wireshark. DNS query and response inspection helped explain name resolution, CNAME and A records, TTL values and response timing. TCP analysis demonstrated the three-way handshake and connection termination, while UDP inspection showed packet fields and UDP-based QUIC traffic. The results improved practical understanding of network protocols and packet-level troubleshooting.

11. Evidence Screenshots

The following screenshots are selected from the supplied Wireshark evidence. Duplicate captures have been omitted. Screenshots support the DNS observations; TCP and UDP observations are documented from the packet analysis previously shared in this conversation.

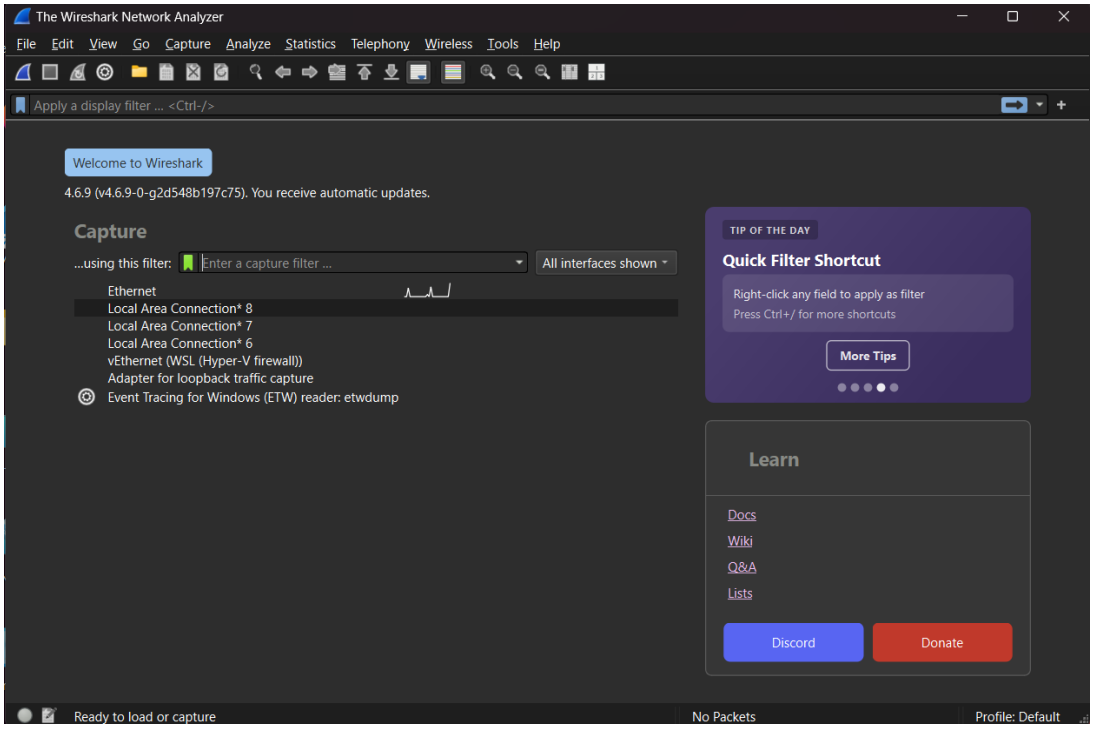


Figure 1. Wireshark 4.6.9 interface showing available capture interfaces.

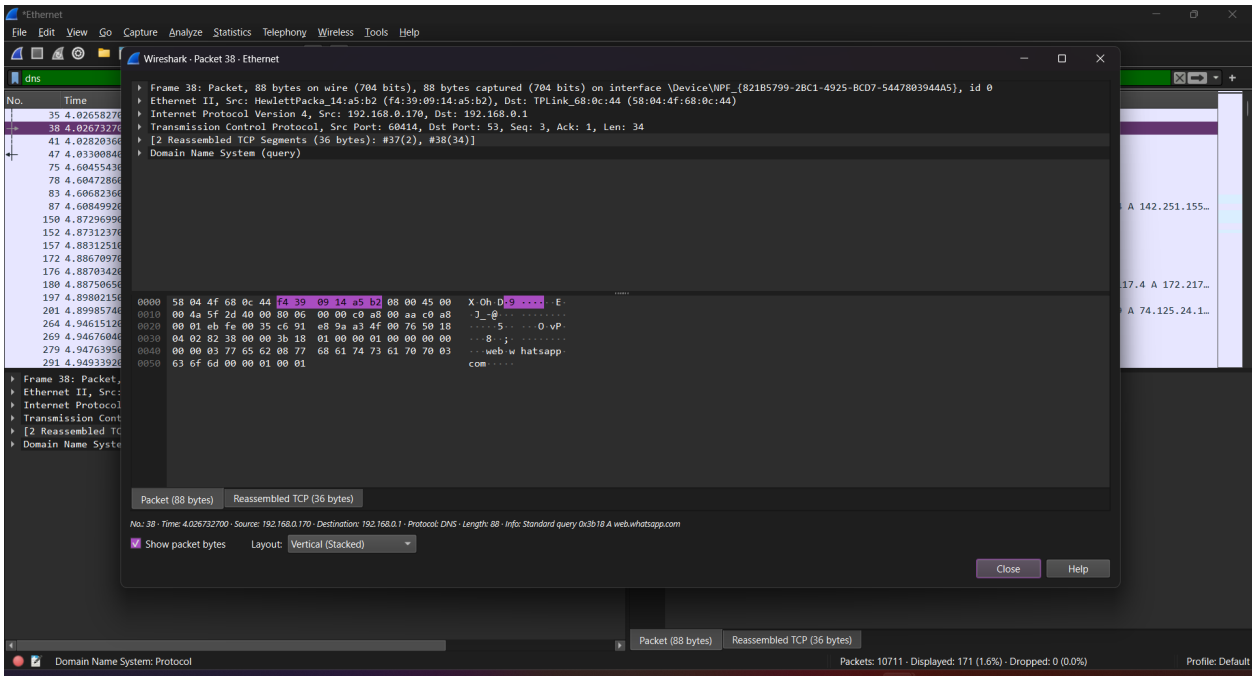


Figure 2. DNS query packet 38 showing the query for web.whatsapp.com.

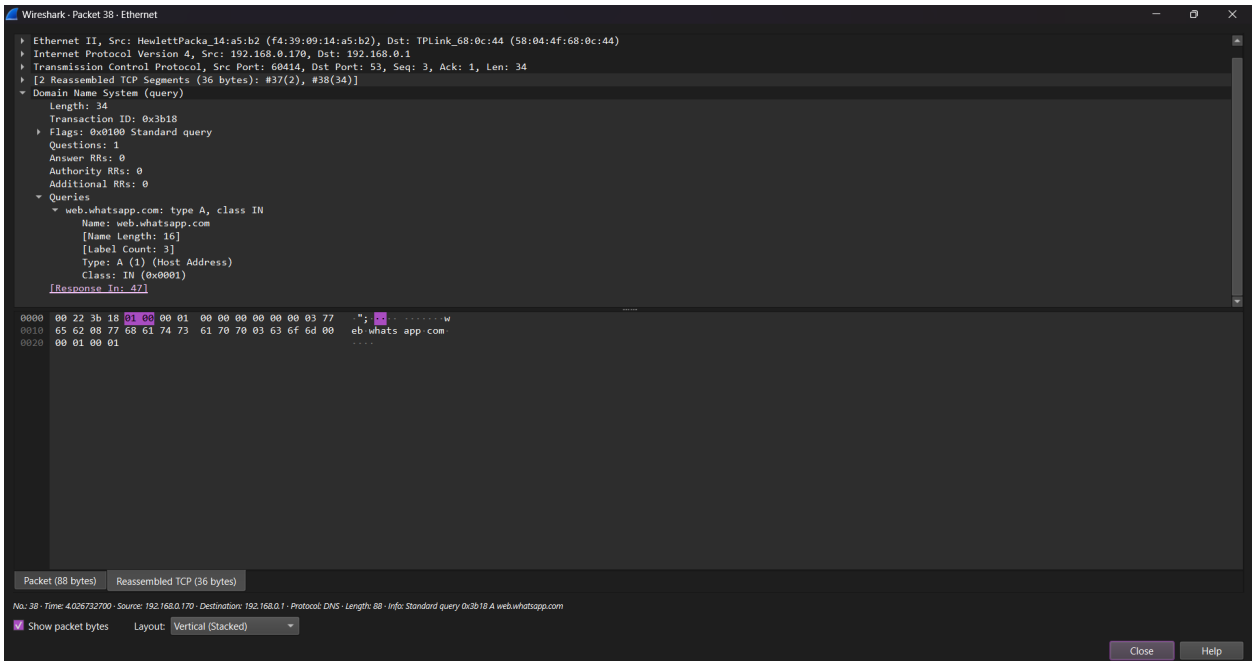


Figure 3. Expanded DNS query details showing the A-record question and linked response packet.

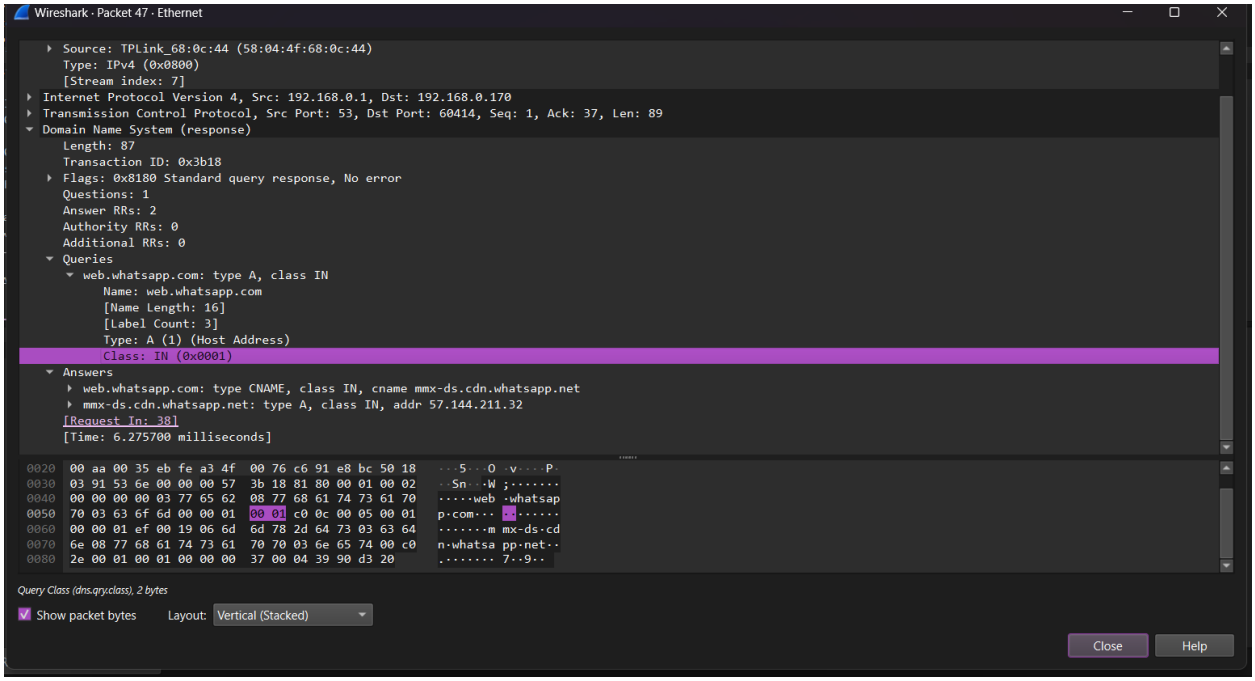


Figure 4. DNS response packet 47 showing successful resolution and response time.

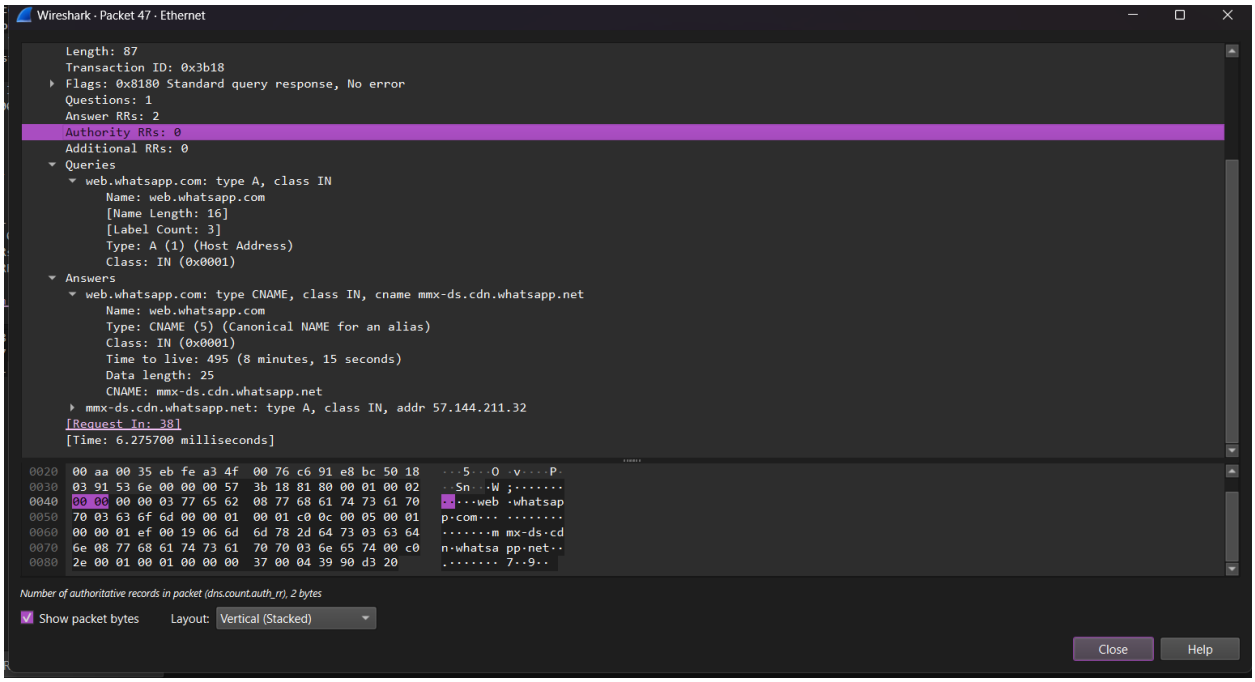


Figure 5. DNS answer details showing the CNAME record and its 495-second TTL.

Evidence limitation: The screenshots supplied for this report primarily document DNS analysis. Separate TCP and UDP screenshots may be added if available. The TCP and UDP values in this report are based on the packet details shared during the analysis conversation.